Assignment 2

Deconstruct, Reconstruct Web Report

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☑ I have read and understood the Declaration and Statement of Authorship above.

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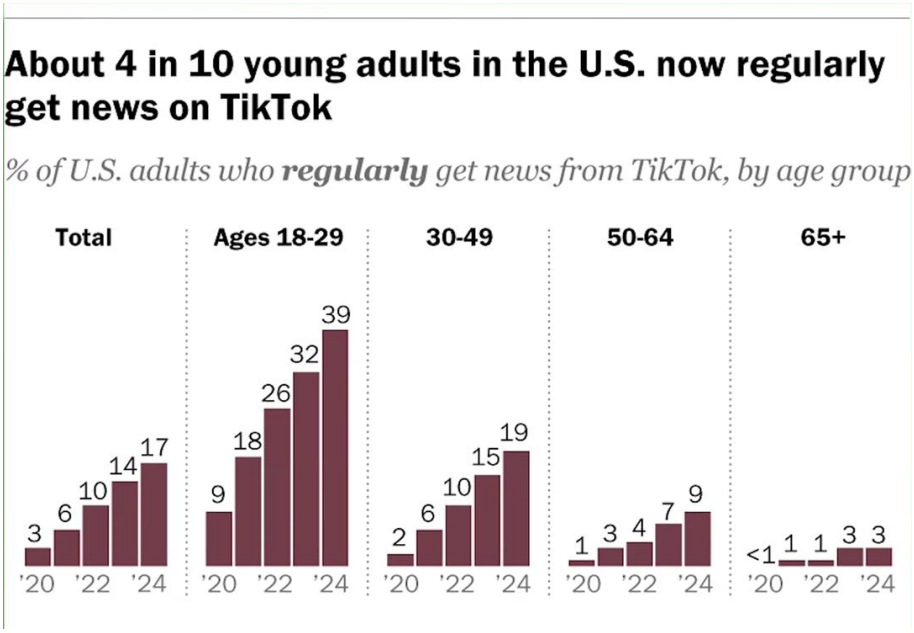
Deconstruct

Original

The original data visualization used is a bar chart from the Pew Research Center that breaks down the percentage of American people who regularly obtain news from TikTok by age group between 2020 and 2024. Each age group is displayed in a separate faceted bar chart using the same maroon color, which makes comparison difficult.



Source: Leppert, R., & Matsa, K. E. (2024, September 17). More Americans – Especially Young Adults – Are Regularly Getting News on TikTok. Pew Research Center. https://www.pewresearch.org/short-reads/2024/09/17/more-americans-regularly-get-news-on-tiktok-especially-young-adults/

Objective and Audience

The objective and audience of the original data visualisation chosen can be summarised as follows:

Objective
This data visualization's goal is to show how, across all age groups, the proportion of American adults who routinely acquire their news from TikTok has evolved between 2020 and 2024. It seeks to draw attention to the growing use of TikTok as a news source by younger audiences, especially those between the ages of 18 and 29.

Audience
The target audience consists of journalists, policymakers, media scholars, and members of the general public, especially those who are curious in changes in the ways that different age groups consume digital media.

Improvements

1. **Combine Facets Into a Unified, Clearer Plot**
   In the original visualization, faceting divided each age group into its own plot. This makes each plot less cluttered, but it makes it more difficult to compare age groups directly from year to year. When all the groups are combined into a single grouped bar chart, the patterns across age groups are easier to understand briefly, and comparative readability is improved.

2. **Add a Clear Legend and Age Group Labels**
   The original graphic identifies age categories simply by subplot titles, which do not include a legend. A single legend enhances clarity and enables readers to rapidly connect colors with certain age groups without depending on the sidebar background. Consistent axis formatting further strengthens interpretability.

3. **Improve Color and Accessibility**
   Users with color vision deficiencies may experience visual problems due to the maroon hue utilized in all aspects, which does not differentiate across age groups.

Better contrast, accessibility, and age-group difference are ensured by switching to a palette like Set2 from the RColorBrewer package.

Reconstruct

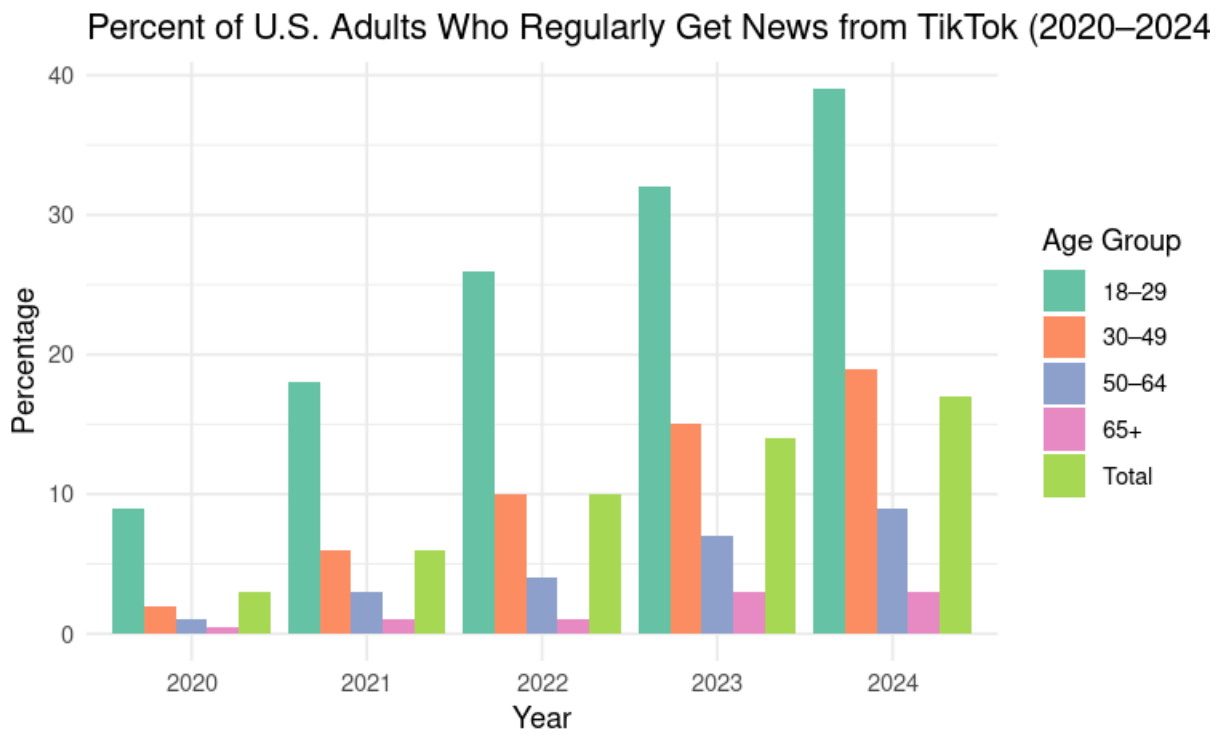
Code

The following code was used to improve the original.

```
library(ggplot2)
library(readxl)
library(dplyr)

# Load the data
data <- read_excel("tiktok_news_by_age.xlsx")

# Reorder age groups
data$Age_Group <- factor(data$Age_Group, levels = c("18–29", "30–49", "50–
64", "65+", "Total"))

# Create grouped bar chart
p1 <- ggplot(data, aes(x = factor(Year), y = Percentage, fill = Age_Group)) +
  geom_col(position = "dodge") +
  labs(
    title = "Percent of U.S. Adults Who Regularly Get News from TikTok (2020–
2024)",
    x = "Year",
    y = "Percentage",
    fill = "Age Group"
  ) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()

p1  # Display the plot
```

Reconstruction

The following plot improves the original data visualisation in the three ways previously explained.



The updated TikTok news consumption chart is found above. Through color choices, it improves accessibility, labeling, and layout clarity.

References

The following references include the original data visualisation selected, the data source(s) used for the reconstruction, and any other sources cited in this report:

Leppert, R., & Matsa, K. E. (2024, September 17). More Americans – Especially Young Adults – Are Regularly Getting News on TikTok. Pew Research Center. https://www.pewresearch.org/short-reads/2024/09/17/more-americans-regularly-get-news-on-tiktok-especially-young-adults/